

* Scalars cannot couple to energy in a Lorentz inv. way

* Focus on low energy theory: do not need UV behavior

• Scales way below the plank mass

$$M_{Pl} \sim 10^{19} \text{ GeV}$$

• Derive expansion for correction terms in reparam order-by-order $\lim_{s \rightarrow 0} \frac{1}{s} (V(s) - V^P(s))$

Review: $DN \circ = s \rightarrow \frac{1}{s} (V(s) - V^P(s))$

s : affine parameter

$$V^P(s) = \phi_* V$$

• pt particle $S = -m \int ds$

$$ds^2 = g_{\mu\nu} dx^\mu dx^\nu$$

via $\delta S = 0$:

$$\begin{aligned} \delta S = 0 &\Rightarrow \int \delta \left(\sqrt{g_{\mu\nu} \dot{x}^\mu \dot{x}^\nu} \right) \quad \text{wrt } g \\ &= \int \frac{1}{2} \left(\right)^{-1/2} \delta g_{\mu\nu} \dot{x}^\mu \dot{x}^\nu \\ &\quad (g_{\mu\nu} \dot{x}^\mu \dot{x}^\nu)^{-1/2} \dot{x}^\mu \dot{x}^\nu \delta g_{\mu\nu} = 0 \end{aligned}$$

$$\begin{aligned} \leadsto \delta \left(-m \int \sqrt{g_{\mu\nu} \dot{x}^\mu \dot{x}^\nu} ds \right) \\ = -m \int \frac{1}{2} (g_{\mu\nu} \dot{x}^\mu \dot{x}^\nu)^{-1/2} \delta g_{\mu\nu} \dot{x}^\mu \dot{x}^\nu \end{aligned}$$

Now

$$\Rightarrow \delta g_{\mu\sigma} g^{\sigma\nu} + g_{\mu\sigma} \delta g^{\sigma\nu} = 0$$

$$\delta g_{\mu\sigma} g^{\sigma\nu} = -g_{\mu\sigma} \delta g^{\sigma\nu}$$

$$\delta g_{\mu\sigma} \delta^\sigma_\lambda = -g_{\mu\sigma} g_{\lambda\nu} \delta g^{\sigma\nu}$$

$$\Rightarrow \delta g_{\mu\lambda} = -g_{\mu\sigma} g_{\lambda\nu} \delta g^{\sigma\nu}$$

$$\therefore -m \int \frac{1}{2} (g_{\mu\nu} \dot{x}^\mu \dot{x}^\nu)^{-1/2} (-g_{\mu\sigma} g_{\lambda\nu} \delta g^{\sigma\lambda}) \dot{x}^\mu \dot{x}^\nu$$

• Variation of x : not of g !

$$-m \int \frac{1}{2} \left(\right)^{-1/2} \delta (g_{\mu\nu} \dot{x}^\mu \dot{x}^\nu)$$

$$(\delta g_{\mu\nu}(x) \dot{x}^\mu \dot{x}^\nu + g_{\mu\nu} \delta \dot{x}^\mu \dot{x}^\nu + g_{\mu\nu} \dot{x}^\mu \delta \dot{x}^\nu) \\ = -m \int \frac{1}{2} (-1)^{-1/2} (\delta g_{\mu\nu} \dot{x}^\mu \dot{x}^\nu + 2 g_{\mu\nu} \dot{x}^\nu \delta \dot{x}^\mu)$$

$$\delta x^\mu = \frac{\partial x^\mu}{\partial s} \delta s$$

$$\delta g_{\mu\nu} = \frac{\partial g_{\mu\nu}}{\partial x^\sigma} \delta x^\sigma = \frac{\partial g_{\mu\nu}}{\partial x^\sigma} \frac{\partial x^\sigma}{\partial s} \delta s$$

• Both parameters

$$\frac{\partial g_{\mu\nu}}{\partial x^\sigma} \delta x^\sigma \dot{x}^\mu \dot{x}^\nu + 2 g_{\mu\nu} \dot{x}^\nu \delta \dot{x}^\mu$$

→

(E. L. Equations)

$$\partial_\mu \frac{\partial \mathcal{L}}{\partial \dot{x}^\mu} - \frac{\partial \mathcal{L}}{\partial x^\mu} = 0$$

$$\frac{\partial \mathcal{L}}{\partial (\partial_\mu x^\sigma)} \quad \mu = t$$

$$\frac{\partial \mathcal{L}}{\partial x^\sigma} = \frac{1}{2} (-1)^{-1/2} 2 g_{\mu\sigma} \dot{x}^\mu$$

$$\frac{\partial \mathcal{L}}{\partial \dot{x}^\sigma} = \frac{1}{2} (-1)^{-1/2} (\partial_\sigma g_{\mu\nu}) \dot{x}^\mu \dot{x}^\nu$$

$$\frac{\partial}{\partial t} \left[(g_{\mu\nu} \dot{x}^\mu \dot{x}^\nu)^{-1/2} g_{\mu\sigma} \dot{x}^\mu \right] \\ = -\frac{1}{2} (g_{\mu\nu} \dot{x}^\mu \dot{x}^\nu)^{-3/2} \partial_t (g_{\mu\nu} \dot{x}^\mu \dot{x}^\nu) g_{\mu\sigma} \dot{x}^\mu \\ + (g_{\mu\nu} \dot{x}^\mu \dot{x}^\nu)^{-1/2} \partial_t (g_{\mu\sigma} \dot{x}^\mu)$$

$$= -\frac{1}{2} (g_{\mu\nu} \dot{x}^\mu \dot{x}^\nu)^{-3/2} \left[(\partial_t g_{\mu\nu}) \dot{x}^\mu \dot{x}^\nu g_{\mu\sigma} \dot{x}^\mu + g_{\mu\nu} 2 \dot{x}^\mu \ddot{x}^\nu g_{\mu\sigma} \dot{x}^\mu \right] \\ + (g_{\mu\nu} \dot{x}^\mu \dot{x}^\nu)^{-1/2} (\partial_t g_{\mu\sigma} \dot{x}^\mu + g_{\mu\sigma} \ddot{x}^\mu)$$

$$\bullet \partial_t g_{\mu\nu} = \dot{x}^\alpha \partial_\alpha g_{\mu\nu}$$

$$\Rightarrow -\frac{1}{2} (g_{\mu\nu} \dot{x}^\mu \dot{x}^\nu)^{-3/2} \left[\dot{x}^\alpha \partial_\alpha g_{\mu\nu} \dot{x}^\mu \dot{x}^\nu + 2 g_{\mu\nu} \dot{x}^\mu \ddot{x}^\nu \right] g_{\mu\sigma} \dot{x}^\mu \\ + (g_{\mu\nu} \dot{x}^\mu \dot{x}^\nu)^{-1/2} (\dot{x}^\alpha \partial_\alpha g_{\mu\sigma} \dot{x}^\mu + g_{\mu\sigma} \ddot{x}^\mu)$$

$$\Rightarrow \boxed{\frac{d^2 x^\alpha}{ds^2} + \Gamma_{\beta\gamma}^\alpha \frac{dx^\beta}{ds} \frac{dx^\gamma}{ds} = 0}$$

(free, massive
particle moving
space-time)

$$\text{Defined by: } D_\mu e_\nu =: \Gamma_{\mu\nu}^\lambda e_\lambda \\ D_\mu A^\nu = \partial_\mu A^\nu + \Gamma_{\mu\alpha}^\nu A^\alpha$$

Weaken field limit: $\int d^4x \sqrt{-g} R$, $g \equiv \det(g_{\mu\nu})$

• $\int d^4x \sqrt{-g}$ is Lorentz invariant.

$$x^\mu \mapsto \Lambda^\mu_\nu x^\nu, \quad J \equiv \det(\Lambda)$$

Proof: $\int d^4x' \sqrt{-g'}$

$$dx'^0 \wedge \dots \wedge dx'^3 = \frac{\partial x'^0}{\partial x^0} dx^0 \wedge \dots \left(\frac{\partial x'^3}{\partial x^3} dx^3 \right)$$

$$= \left(\frac{\partial x'^0}{\partial x^0} \dots \frac{\partial x'^3}{\partial x^3} \right) dx^0 \wedge \dots \wedge dx^3$$

$$= \frac{\partial x'^0}{\partial x^0} \dots \frac{\partial x'^3}{\partial x^3} \delta^0_{\alpha_0} \dots \delta^3_{\alpha_3} dx^{\alpha_0} \wedge \dots \wedge dx^{\alpha_3}$$

$$= \left(\frac{\partial x'^0}{\partial x^{\alpha_0}} \dots \frac{\partial x'^3}{\partial x^{\alpha_3}} \right) \delta^{\alpha_0 \dots \alpha_3}_{0 \dots 3} dx^{\alpha_0} \wedge \dots \wedge dx^{\alpha_3}$$

$$= (1) \delta^0 \dots 3 dx^0 \wedge \dots \wedge dx^3$$

$$= \left(\frac{\partial x'^0}{\partial x^{\alpha_0}} \dots \frac{\partial x'^3}{\partial x^{\alpha_3}} \right) \epsilon^{\alpha_0 \dots \alpha_3} dx^{\alpha_0} \wedge \dots \wedge dx^{\alpha_3}$$

$$= \det(\Lambda) dx^0 \wedge \dots \wedge dx^3 = J dx^0 \wedge \dots \wedge dx^3$$

$$g' = \det(g'_{\mu\nu})$$

$$g_{\mu\nu} \rightarrow \Lambda^\alpha_\mu g_{\alpha\beta} \Lambda^\beta_\nu = g'_{\mu\nu}$$

$$\det(g') = \det(g) (\det(\Lambda))^{-2}$$

and so the measure is invariant

• Promote Λ to a local symmetry:

$$x^\mu \mapsto x'^\mu = \Lambda^\mu_\nu(x) x^\nu$$

$$dx^\mu \mapsto dx'^\mu = \Lambda^\mu_\nu(x) dx^\nu$$

and so

$$\Lambda^\mu_\nu(x) = \delta^\mu_\nu + e^\mu_\nu(x) + \mathcal{O}(e^2)$$

$$\Rightarrow x'^\mu = (\delta^\mu_\nu + e^\mu_\nu(x)) x^\nu$$

$$= x^\mu + e^\mu_\nu(x) x^\nu$$

• define $E^\mu(x) \equiv x^\nu e^\mu_\nu(x)$

• $[D_\mu, D_\nu] = F_{\mu\nu}$, here

$$[D_\mu, D_\nu] A_\alpha \equiv R^\beta_{\alpha\mu\nu} A_\beta$$

$$R^\beta_{\alpha\mu\nu} = \partial_\mu \Gamma^\beta_{\alpha\nu} - \partial_\nu \Gamma^\beta_{\alpha\mu} + \Gamma^\lambda_{\alpha\mu} \Gamma^\beta_{\lambda\nu} - \Gamma^\lambda_{\alpha\nu} \Gamma^\beta_{\lambda\mu}$$

$$R_{\mu\nu} \equiv R^\alpha_{\mu\nu\alpha}$$

$$R \equiv g^{\mu\nu} R_{\mu\nu}$$

* Weyl Curvature tensor $C^\alpha_{\mu\nu\rho}$: trace free

$$S_{EH} = \frac{2}{\kappa^2} \int d^4x \sqrt{-g} \mathcal{L}, \quad \kappa = \frac{1}{M_{Pl}}$$

$$R_{\mu\nu} - \frac{1}{2} g_{\mu\nu} R = 8\pi G_N T_{\mu\nu}$$

• Energy of the gravitating body, not its mass, that defines the curvature of S.T. around it

• Background field method:

Introduce a background field $\bar{g}_{\mu\nu}$

$$g_{\mu\nu} = \bar{g}_{\mu\nu} + \kappa h_{\mu\nu}$$

* Since we are only keeping $\mathcal{O}(\kappa^2)$ we can raise and lower the above using only the background field

$$g_{\mu\alpha} g^{\alpha\nu} = \delta^\nu_\mu$$

$$(\bar{g}_{\mu\alpha} + \kappa h_{\mu\alpha}) g^{\alpha\nu} = \delta^\nu_\mu$$

$$g_{\mu\nu} = \bar{g}_{\mu\nu} + \kappa h_{\mu\nu}$$

$$g^{\alpha\beta} = \bar{g}^{\alpha\beta} - \kappa h^{\alpha\beta} + \kappa^2 h^\alpha_\alpha h^{\beta\beta}$$

• Can be defined with the same metric
 and take $g_{\mu\nu} = \eta_{\mu\nu}$ of $M^{1,3}$
 Then $h_{\mu\nu}$ transforms as a gauge transformation
 under int. diffeomorphisms
 $x^\mu \rightarrow x'^\mu + \epsilon^\mu(x) = x'^\mu$

$$h_{\mu\nu}(x) \rightarrow h'_{\mu\nu}(x) = \Lambda^\alpha_\mu(x) h_{\alpha\beta}(x) \Lambda^\beta_\nu(x)$$

$$\begin{aligned} h'_{\mu\nu}(x) &= \Lambda^\alpha_\mu(x) h_{\alpha\beta}(x) \Lambda^\beta_\nu(x) \\ &= (\delta^\alpha_\mu + \epsilon^\alpha_\mu) h_{\alpha\beta} (\delta^\beta_\nu + \epsilon^\beta_\nu) \\ &= h_{\mu\nu} + \epsilon^\alpha_\mu h_{\alpha\beta} + \epsilon^\beta_\nu h_{\alpha\beta} \\ &= h_{\mu\nu} + \epsilon^\alpha_\mu h_{\alpha\beta} + \epsilon^\alpha_\nu h_{\alpha\beta} \\ &= h_{\mu\nu} + (\epsilon^\alpha_\mu + \epsilon^\alpha_\nu) h_{\alpha\beta} \end{aligned}$$

$$\Rightarrow h_{\mu\nu}(x + \epsilon) = h(x) + \epsilon^\alpha \partial_\alpha h_{\mu\nu}$$

$$\begin{aligned} h'_{\mu\nu}(x') &= \Lambda^\alpha_\mu(x') h_{\alpha\beta}(x') \Lambda^\beta_\nu(x') \\ &= (\Lambda(x) + x^\lambda \partial_\lambda \Lambda) h_{\alpha\beta} \Lambda^\beta_\nu(x') \end{aligned}$$

we find that

$$h'_{\mu\nu}(x) = h_{\mu\nu}(x) - \partial_\mu \epsilon_\nu(x) - \partial_\nu \epsilon_\mu(x)$$

* Express it as:

$$\sqrt{-g} = \sqrt{-\det(\bar{g}_{\mu\nu} + \kappa h_{\mu\nu})}$$

$$\log \det(g) = \text{Tr}(\log(\bar{g} + \kappa h_{\mu\nu}))$$

$$\begin{aligned} &\text{Tr}(\log(\bar{g}^{-1}(\mathbb{1} + \kappa \bar{g}^{-1} h_{\mu\nu}))) \\ &= \text{Tr}(\log(\bar{g}^{-1})) + \text{Tr}(\log(\mathbb{1} + \kappa \bar{g}^{-1} h_{\mu\nu})) \\ &= -\frac{1}{2} \text{Tr}(\log(\bar{g})) + \text{Tr}(\log(\mathbb{1} + \kappa \bar{g}^{-1} h_{\mu\nu})) \end{aligned}$$

$$= \text{Tr}(\log(g)) + \text{Tr}(\log(1 + \kappa g^{-1} h_{\mu\nu}))$$

$$\log(1+x) = \sum_{n=0}^{\infty} \frac{1}{n} (-1)^n x^n$$

$$\log(1 + \kappa \bar{g}^{-1} h_{\mu\nu}) = \sum_{n=0}^{\infty} \frac{1}{n} (-1)^n (\bar{g}^{\mu\nu} h_{\mu\nu})^n$$

$$\det(g) = e^{\text{Tr}(\log(\bar{g}))} e^{\sum_n \frac{1}{n} (-1)^n (\bar{g}^{\mu\nu} h_{\mu\nu})^n}$$

$$= \det(\bar{g}) e^{\sum_n \frac{1}{n} (-1)^n (\bar{g}^{\mu\nu} h_{\mu\nu})^n}$$

$$\Rightarrow h \equiv \bar{g}^{\mu\nu} h_{\mu\nu}$$

$$\det(g) = \det(\bar{g}) e^{\sum_n \frac{1}{n} (-1)^n h^n}$$

$$\sqrt{g} = 1 + \frac{1}{2} \kappa h - \frac{\kappa^2}{4} h_{\alpha\beta} \rho^{\alpha\beta, \mu\nu} h_{\mu\nu} + \mathcal{O}(\kappa^3)$$

Up to linear order $\mathcal{O}(\kappa)$ the Riemann tensor takes up the form:

$$\Gamma_{\mu\nu}^{\lambda} = \frac{1}{2} (\partial_{\mu} h_{\nu}^{\lambda} + \partial_{\nu} h_{\mu}^{\lambda} - \partial^{\lambda} h_{\mu\nu})$$

which distributes g past ∂_{μ} for some reason...

$$R = \square h - \partial_{\lambda} \partial_{\sigma} h^{\lambda\sigma}$$

$$\square h_{\mu\nu} = -16\pi G (T_{\mu\nu} - \frac{1}{2} \eta_{\mu\nu} T^{\alpha}_{\alpha})$$